

Effect of chemical composition of ash on sintering of lignites in circulating fluid bed combustion and successful operation of large CFBC boilers

The original paper [↗](#) contains 33 sections, with 10 passages identified by our machine learning algorithms as central to this paper.

Paper Summary

SUMMARY PASSAGE 1

Backdrop/Introduction

By judiciously combining the advanced analytical & empirical techniques we can pin point the impurity, mineral and its transformation causing the sintering/agglomeration. The present research focuses to understand the mechanism of sintering of fly ashes further through a study of the chemical composition of lignite resource at mines, the mineral analysis/ mineralogy of lignite samples by advanced analytical techniques in addition to morphology of sintered fly ash and presents a useful predictive correlation based on the ash chemical composition. This correlation based on commonly used coal ash chemical composition in the assessment of quantum of sintering would save considerable time, effort and costs for the utilities.

SUMMARY PASSAGE 2

Sintering Characterization Using The Heating Microscope

Various ash sintering measurement techniques, growth measurements between spherical particles using microscope, simultaneous linear shrinkage and electrical conductance measurements, crushing/compressive strength measurements of sintered ash pellets, ash agglomeration by sieving tests and the area shrinkage measurement method were reviewed recently [3,4]. The sintering behavior of lignite coal ashes in the temperature range of 800 to 1000 C for CFBC applications was demonstrated using heating microscopy. The characteristic temperatures-Initial sintering Temperature ITs (temperature at which sintering/shrinkage starts), Peak sintering Temperature PTs (temperature at which the shrinkage rate is maximum) and final sintering Temperature CTs (temperature at which no significant shrinkage), and the maximum rate of shrinkage percent per degree Celsius (R max) were studied.

SUMMARY PASSAGE 3

Computer Controlled Scanning Electron Microscope

The conventional tests ash fusion temperature and chemical composition of ash are not utilized convincingly for identifying sintering related problems in CFBC [3]. Given the shortcomings of the empirical indices, it would seem likely that using better analytical data about the mineral matter in coal and with a better understanding of transformation mechanisms, the prospects for prediction will improve [7]. Computer Controlled Scanning Electron Microscope (CCSEM) is a widely used automated technique used to determine the size, association, composition and abundance of minerals in coal on a particle-by-particle basis [8].

SUMMARY PASSAGE 4

Experimental Procedure Laboratory

The Å72 mesh samples are used for analyses of proximate, ultimate and calorific value. Adequate quantity of ash of each fuel is generated as per proximate analyses at 750 C for further analyses of chemical composition, ash fusion temperature and for sintering measurement. The proximate, ultimate and gross calorific value of the samples were carried out using TGA 701 proximate analyzer (LECO), elemental analyzer Vario EL III and PARR Isoperibol bomb calorimeter respectively.

SUMMARY PASSAGE 5

Area Shrinkage Curves And Sintering Profiles

Two test runs were carried out for each lignite sample with accuracy and reproducibility as per as per DIN 51730 (1998e4). The area shrinkage percentage due to sintering of the ash pellet vs. temperature was plotted for all the ash samples. Sintering measurement and ash fusion tests were carried out by Leica heating microscope (Leica Microsystems, Wetzlar, Germany).

SUMMARY PASSAGE 6

Modifications In Operating Procedure To Overcome Sintering Related Clogging

High sintering was noticed during compression strength tests in case of lignite ash at 900 C whereas cyclone deposit did not undergo any sintering. But 30% by weight lignite ash addition to the cyclone deposit resulted in formation of sinters at 900 C. The ash particles, when exposed to higher temperature during ashing, undergoes some amount of sintering and to that extent, shrinkage pattern either gets reduced during shrinkage measurement and/or shifts to higher temperature. This phenomenon i.e., with the higher ashing temperature, the lignite ash loses part of its sintering tendency which indicated to a hypothesis that once the lignite ash undergoes transformations resulting in sintering, its sintering tendency is lowered [3].

SUMMARY PASSAGE 7

Chemical Composition Of The Coal And Lignite

However, it is experienced that firing of the coal samples 1 and 9 resulted in clogging/blockage during initial operation of the CFB boilers (Fig. 1). It shall be noted, agglomerates are formed during operation, several hundred degrees below the initial deformation temperatures of coal ash tested by ASTM procedure. With respect to sintering property of coal ash, the ash fusion test (AFT) tends to be rather imprecise and subjective [14].

SUMMARY PASSAGE 8

Sintering Pattern Of Samples

The area shrinkage percent due to sintering of the ash pellet vs. temperature is plotted for all the ten ash samples in Fig. 4. The derivative of shrinkage i.e., rate of area shrinkage dA/dT is also calculated for different level of temperature for all the ten ash samples and are shown in Fig. 4.

SUMMARY PASSAGE 9

Influence Of Chemical Composition In Sintering

The uniqueness of samples 1- & 9 that could be noticed is that the $\text{SiO}_2 \sim \frac{3}{4} \text{Al}_2\text{O}_3$ percentage in the total ash is more than 80%; and the SiO_2 to Al_2O_3 ratio is between 1.0 and 1.25 only for samples 1 & 9 as seen in Table 1. The {ratio of Silica to Alumina} and the {sum total of % Silica, and % Alumina} by weight in the ash are the factors for classifying the lignite ash for sintering phenomena. Based on numerous field trials and operating experience with sintering lignite and the laboratory sample results the lignites were grouped as below in three distinct groups [3].

SUMMARY PASSAGE 10

Conclusions

Widely varying sintering behavior of these coals, due to the different chemical composition of ash, are characterized based on the silica alumina ratio and a useful thumb rule correlation is derived, to predict total quantum of sintering, identify that would cause severe agglomeration in advance. The operating personnel, whenever faced with fuels that would cause heavy sintering, as predicted by the empirical equation, can select suitable optimum temperature range of combustion to achieve "dead burning" of ash in the combustor. A novel operational method, that combines BFB mode and CFB mode, during varying loads, to achieve optimum temperature range by process modification that overcame the sintering related clogging and the unscheduled shutdown of the plant, was successfully developed and demonstrated.